3rd Year Project, Heating System Design
Air Heater

Air Heat Exchanger

A fin and tube design, for heating the air shell side, using the heat from the transfer fluid, tube side. Base equations taken from Heat Exchanger Design Coursework from semester 1.

```
> restart;
> #Equations
  R := (T__hot1 - T__hot2)/(T__cold2 - T__cold1);
  S := (T__cold2 - T__cold1)/(T__hot1 - T__cold1);
  A__req := Q/(F*U*DeltaT__LM);
  DeltaT__LM := ((T__hot2 - T__cold1)-(T__hot1 - T__cold2))/ln(
  (T__hot2-T__cold1)/(T__hot1-T__cold2));
  T__hot2 := T__hot1 - Q/(m__hot*c__hot); #Outlet temperature of
  hot fluid
```

$$R := \frac{T_{hot1} - T_{hot2}}{T_{cold2} - T_{cold1}}$$

$$S := \frac{T_{cold2} - T_{cold1}}{T_{hot1} - T_{cold1}}$$

$$A_{req} := \frac{Q}{F\, U\, DeltaT_{LM}}$$

$$DeltaT_{LM} := \frac{T_{hot2} - T_{cold1} - T_{hot1} + T_{cold2}}{\ln\left(\frac{T_{hot2} - T_{cold1}}{T_{hot1} - T_{cold2}}\right)}$$

$$T_{hot2} := T_{hot1} - \frac{Q}{m_{hot}\, c_{hot}} \tag{1.1}$$

```
> #1 is in
  #2 is out
  #Cold is air
  #Hot is oil/salt/water

  print("====Specifications====");
  m__cold := 0.0570722222;#kg/s      (Air mass)
```

```
T__cold2 := 280 + 273.15;#Kelvin  (Air Temp)
m__hot := 0.25;#kg/s        #Okay for now, but can be varied (Hot
fluid mass flow)

T__cold1 := 20 + 273.15; #Kelvin (Ambient Air Temperature)


T__hot1 := 330 + 273.15; #Kelvin
c__hot := 2150; #J/kg/K (Isobaric)   [Same as cp]
c__cold := 1014; #J/kg/K (Isobaric)  [Same as cp]
Q := m__cold*c__cold*(T__cold2 - T__cold1); #W

print ("====Properties of Helisol 5A, tube side====");
rho__t := 689.5; #kg/m3         #
mu__t := 0.41*10^(-3); #Pa*s
k__t := 0.076; #W/m/K
C__pt := 2150; #J/kg/K

print("=====Estimated=====");
U__o_ass := 3.793; #W/m2/K
```

"====Specifications===="

$$m_{cold} := 0.0570722222$$

$$T_{cold2} := 553.15$$

$$m_{hot} := 0.25$$

$$T_{cold1} := 293.15$$

$$T_{hot1} := 603.15$$

$$c_{hot} := 2150$$

$$c_{cold} := 1014$$

$$Q := 15046.52066$$

"====Properties of Helisol 5A, tube side===="

$$\rho_t := 689.5$$

$$\mu_t := 0.0004100000000$$

$$k_t := 0.076$$

$$C_{pt} := 2150$$

"=====Estimated====="

$$U_{o\_ass} := 3.793 \quad \textbf{(1.2)}$$

```
> R := R;
  S := S;
  A__req;
  T__hot2;
  T__cold2;
```

$$R := 0.1076674108$$

$$S := 0.8387096774$$

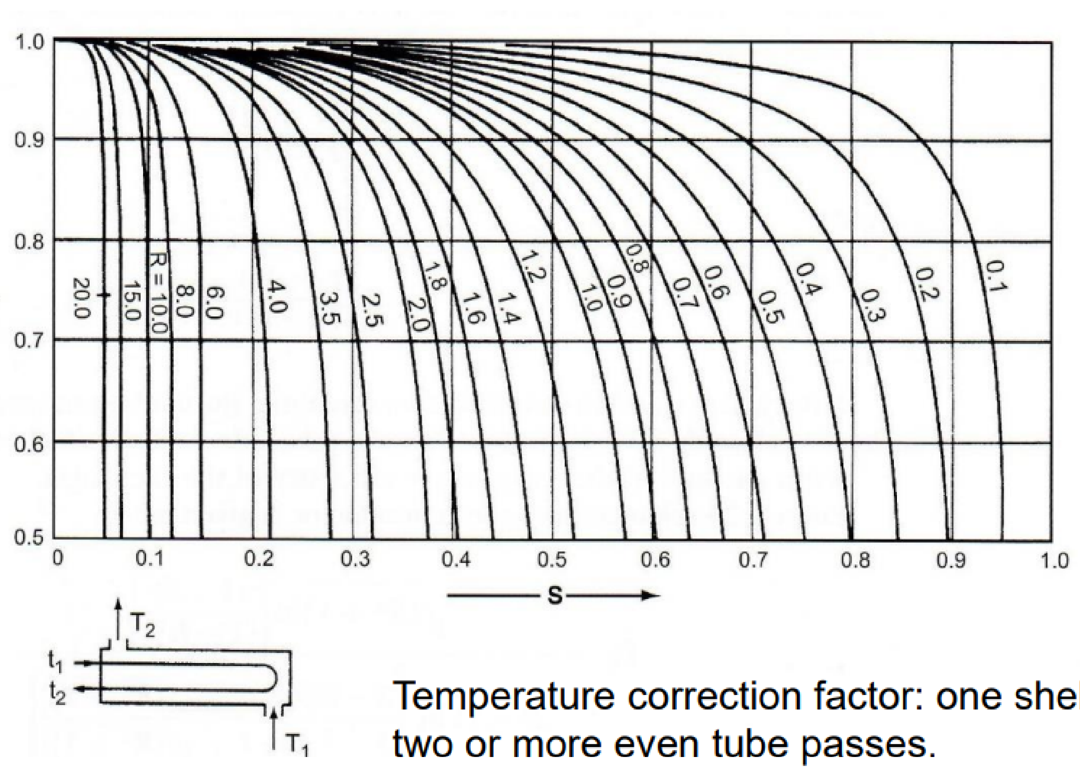
$$\frac{112.1911874}{F\,U}$$

$$575.1564732$$

$$553.15 \quad \textbf{(1.3)}$$

```
>
```

The “correction factor” *F*



Temperature correction factor: one shell pass; two or more even tube passes.

```
> F := 0.95; #VERY ROUGH, NEED TO MEASURE PROPERLY
```

$$F := 0.95 \quad \textbf{(1.4)}$$

```
> U := U__o_ass; #TEMPORARY ASSUMPTION
```

```
A__req; #m2
DeltaT__LM;
```

$$U := 3.793$$

$$31.13524564$$

$$134.1149961 \qquad (1.5)$$

```
> L__tube := 1.750; #m
D__internal := 0.015; #m
D__external := 0.018; #m
#Fin type, Helical
l__f := 0.010; #m
t__f := 0.0005; #m
p__f := 0.002; #m
r__f1 := D__external/2;
l__fc := l__f + t__f/2; #corrected fin hight to account for
transfer in tip
r__f2c := r__f1 + l__fc;
A__f := 2*Pi*(r__f2c^2 - r__f1^2);
N__fin := L__tube/p__f;
A__b := Pi*D__external*(L__tube - N__fin*t__f);
A__o_one_tube := A__f*N__fin + A__b;
N__tt := A__req/A__o_one_tube;
```

$$L_{tube} := 1.750$$

$$\mathrm{D}_{internal} := 0.015$$

$$\mathrm{D}_{external} := 0.018$$

$$l_f := 0.010$$

$$t_f := 0.0005$$

$$p_f := 0.002$$

$$r_{f1} := 0.009000000000$$

$$l_{fc} := 0.01025000000$$

$$r_{f2c} := 0.01925000000$$

$$A_f := 0.001819374846$$

$$N_{fin} := 875.0000000$$

$$A_b := 0.07422012645$$

$$A_{o\_one\_tube} := 1.666173116$$

$$N_{tt} := 18.68668108 \tag{1.6}$$

```
> #Chosen Layout is a triangular with 30 degrees rotation.
  #Number of passes is 2
  N__p := 2;
  psi__n := 0.17;
  C__1 := 0.866;
  L__tp := 1.25 * (D__external + 2*l__f); #Minimum TEMA standars is
  1.25 https://www.thermopedia.com/content/1121/
```

$$N_p := 2$$

$$\psi_n := 0.17$$

$$C_1 := 0.866$$

$$L_{tp} := 0.04750 \tag{1.7}$$

```
>
  D__ctl := L__tp*(4*C__1*N__tt/3.14159265/(1-psi__n))^(1/2);
```

$$\mathrm{D}_{ctl} := 0.2366653381 \tag{1.8}$$

```
> L__bb := 0.0127; #Baffle bypass clearance
```

$$L_{bb} := 0.0127 \tag{1.9}$$

```
>
  D__s := D__ctl + L__bb + D__external;
  L__B := 3.0*D__s; #Baffle spacing
```

$$\mathrm{D}_s := 0.2673653381$$

$$L_B := 0.8020960143 \tag{1.10}$$

```
> P__v := L__tp * cos(3.14159265/6);
  N__r := D__ctl / P__v;
  BaffleCut := 0.25; #%
  N__tcc := (1-BaffleCut)*N__r;
```

$$P_v := 0.04113620669$$

$$N_r := 5.753212490$$

$$BaffleCut := 0.25$$

$$N_{tcc} := 4.314909368 \tag{1.11}$$

```
> T__coldAvg := (T__cold1 + T__cold2)/2;
```

$$T_{coldAvg} := 423.1500000 \tag{1.12}$$

```
> #Calculating Tube side cross-sectional flow area
  N__per_pass := N__tt/N__p;
  A__internal := (3.14159/4)*D__internal^2; #m2
  A__t := N__per_pass * A__internal; #m2
```

$$N_{per\_pass} := 9.343340540$$

$$A_{internal} := 0.0001767144375$$

$$A_t := 0.001651103168 \qquad \textbf{(1.13)}$$

```
> #Helisol 5A key equations
  v__t := m__hot/(rho__t*A__t);
  Re__t := (rho__t*v__t*D__internal)/mu__t;
  Pr__t := (C__pt*mu__t)/k__t;
  mu__corr := 1;
  Nu__t := 0.023*Re__t^0.8*Pr__t^0.3*(mu__corr); #Adjusted from 0.3
  from 0.4 to account for different boundary Layer Behavior
  h__i := (Nu__t *k__t)/D__internal; #W/m^2/K
```

$$v_t := 0.2195995913$$

$$\Re_t := 5539.533593$$

$$Pr_t := 11.59868421$$

$$\mu_{corr} := 1$$

$$\mathrm{N}_t := 47.40553078$$

$$h_i := 240.1880226 \qquad \textbf{(1.14)}$$

```
> #properties of air, shell side
  rho__s := 0.835; #kg/m3 (air at 280C)
  mu__s := 2.38e-5; #kg/m/s
  k__s := 0.0354; #W/m/K
  C__ps := 1014;
```

$$\rho_s := 0.835$$

$$\mu_s := 0.0000238$$

$$k_s := 0.0354$$

$$C_{ps} := 1014 \qquad \textbf{(1.15)}$$

```
> Q__air := m__cold/rho__s;
  A__face := L__tube*(D__ctl + D__external);
  u__f := Q__air/A__face;
  u__max := u__f*(L__tp/(L__tp - D__external));
```

$$Q_{air} := 0.06834996671$$

$$A_{face} := 0.4456643417$$

$$u_f := 0.1533664696$$

$$u_{\max} := 0.2469460104 \tag{1.16}$$

```
> Pr__s := C__ps*mu__s/k__s;
  Re__s := rho__s * u__max * D__external/mu__s;
  Nu__s := 0.134*Re__s^0.681*Pr__s^0.33*((p__f-t__f)/l__f)^0.2*
  (p__f/t__f)^0.1134;
  h__s := Nu__s * k__s/D__external;
```

$$Pr_s := 0.6817288136$$

$$\Re_s := 155.9495184$$

$$\mathrm{N}_s := 2.945171758$$

$$h_s := 5.792171122 \tag{1.17}$$

```
> #Defining Thermal Conductivity Variables
  k__tube := 50; #W/m/K
  k__fin := 205; #W/m/K
  Rf__o := 0.0002; #m2*K/W
  Rf__i := 0.0003; #m2*K/W
```

$$k_{tube} := 50$$

$$k_{fin} := 205$$

$$Rf_o := 0.0002$$

$$Rf_i := 0.0003 \tag{1.18}$$

```
> #Fin equations
  m := simplify(((2*h__s)/(k__fin * t__f))^(1/2));
  eta__f := tanh(m * l__f) / (m * l__f);
  N__fin := L__tube/p__f;
```

$$m := 10.63099116$$

$$\eta_f := 0.9962496874$$

$$N_{fin} := 875.0000000 \tag{1.19}$$

```
> #Outside area calculations
  A__f := A__f;
  A__b := A__b;
  A__f_total := N__fin * A__f * N__tt;
```

```
A__b_total := A__b * N__tt;
A__o := A__f_total + A__b_total;
```

$$A_f := 0.001819374846$$

$$A_b := 0.07422012645$$

$$A_{f\_total} := 29.74831782$$

$$A_{b\_total} := 1.386927833$$

$$A_o := 31.13524565 \quad (1.20)$$

```
> #Internal Area calculations
A__i := 3.14159 * D__internal * L__tube * N__tt;
```

$$A_i := 1.541029624 \quad (1.21)$$

```
> eta__o := (eta__f *  A__f_total + A__b_total)/A__o
```

$$\eta_o := 0.9964167461 \quad (1.22)$$

```
> #Calculating all thermal resistances
R__shell_conv := 1/eta__o/h__s; #Air side convection
R__tube_conv := 1/(h__i*(A__i/A__o)); #Water side convection
R__wall_cond := (D__external * ln(D__external/D__internal))/(2*
k__tube);
R__fouling := (Rf__i * (A__o/A__i))+(Rf__o /eta__o);
```

$$R_{shell\_conv} := 0.1732676951$$

$$R_{tube\_conv} := 0.08411819835$$

$$R_{wall\_cond} := 0.00003281788022$$

$$R_{fouling} := 0.006261974346 \quad (1.23)$$

```
> #Calculating U
R__total := R__shell_conv + R__tube_conv + R__wall_cond +
R__fouling;
```

$$R_{total} := 0.2636806856 \quad (1.24)$$

```
> print("============================");
U__o_calc := 1/R__total; #W/m2/K
print("============================");
```

"============================"

$$U_{o\_calc} := 3.792465867$$

"============================" (1.25)

```
> #Tube side pressure drop
```

```
f__t := 0.0035 + 0.264/Re__t^0.42;
L__total := L__tube * N__p;
K := 1.8*N__p;
DP__t_friction := 4*f__t*(L__total/D__internal)*(rho__t*v__t^2/2)
;
DP__t_return := K*(rho__t*v__t^2)/2;
DP__t := DP__t_friction + DP__t_return;
```

$$f_t := 0.01056879168$$

$$L_{total} := 3.500$$

$$K := 3.6$$

$$DP_{t\_friction} := 163.9945609$$

$$DP_{t\_return} := 59.85078220$$

$$DP_t := 223.8453431 \tag{1.26}$$

```
> #Shell side pressure drop
A__s := (L__tp - D__external)*D__s*L__B/L__tp;
G__s := m__hot/A__s;
u__s := G__s/rho__s;
d__e := 1.10/D__external*(L__tp^2 - 0.917*D__external^2);
Re__s := G__s*d__e/mu__s;
```

$$A_s := 0.1331863963$$

$$G_s := 1.877068582$$

$$u_s := 2.247986326$$

$$d_e := 0.1197253444$$

$$\Re_s := 9442.549681 \tag{1.27}$$

```
> #Calcualted via Robinson & Briggs

f := 9.465 * Re__s^(-0.316) * (L__tp/(D__external + 2*l__f))^
(-0.927)
```

$$f := 0.4267335849 \tag{1.28}$$

```
> DP__s := N__r * f * (rho__s * u__max^2 / 2)
```

$$DP_s := 0.06250686920 \tag{1.29}$$

```
> print("=====summary=====");
Re__t := Re__t;
Re__s := Re__s;
```

```
DP__t := DP__t; #Pa
DP__s := DP__s; #Pa
U__o_calc := U__o_calc;
U__o_ass := U__o_ass;
```

"=====summary====="

$$\Re_t := 5539.533593$$

$$\Re_s := 9442.549681$$

$$DP_t := 223.8453431$$

$$DP_s := 0.06250686920$$

$$U_{o\_calc} := 3.792465867$$

$$U_{o\_ass} := 3.793 \tag{1.30}$$